F1 Podium Prediction — Milestone 1 Report

Team 54

Team members (name — student ID)

• Ziad Ahmed Ibrahim — 58-0690

• Mahmoud Abdelrahman — 58-1636

• Karim Ahmed Raafat — 58-2591

• Mahmoud Mohamed Abdelgawad — 58-5983

Kaggle notebook and GitHub links are submitted separately on the course form.

Summary

We predict, for every driver in every race, whether they finish on the podium (top 3), using only what is known after qualifying and before the start. This report covers the data audit and cleaning, the three data-engineering questions (DEQs), the leakage-safe feature set, the models and their evaluation, the experiments (including the feature-group ablation), the explanations of the final model, the inference function and the known limitations. Every number comes from the run-all notebook.

Headline answers

- **DEQ1, front-row conversion by circuit.** From 2014 on, Monaco (89.5%), Suzuka (88.9%) and Yas Marina (86.4%) turn a front-row start into a podium most often, and the Hungaroring (59.1%) least often. The ranking changes a lot after 2014: the circuits' 1950–2013 and 2014–2024 rates are uncorrelated (Spearman ρ = −0.11). Over the same period, front-row conversion rose everywhere, from 56.5% to 76.2%. Grid position, not qualifying position, carries the signal: front-row qualifiers who started further back converted only 26.8%.
- **DEQ2, home advantage.** No. Raw home podium rates are *lower* (11.9% vs 13.9%) because home drivers start further back. Once grid position is controlled for, the odds ratio is 0.97 [0.81, 1.15], with no evidence of a home effect.
- **DEQ3, mechanical retirements.** In 2014–2021 the highest rates were Lotus F1 (20.0%), Toro Rosso (16.1%) and Renault (14.1%). In 2022–2024 they were Alfa Romeo (12.5%), Alpine (10.4%) and AlphaTauri (8.0%). The field-wide rate fell from 9.6% to 5.9% per start. The Enstone team (Lotus F1 → Renault → Alpine) is the least reliable lineage in both periods.
- **Model.** The final model is a tuned **logistic regression** on 27 pre-race features. It reaches **PR-AUC 0.728 on validation (2020–21) and 0.711 on the unseen test seasons (2022–24)**, against a base

rate of 0.151 (a grid-position-only rule scores 0.649 and 0.647). Random forest, gradient boosting and a shallow neural network are statistically tied with it (every 95% interval of the difference contains 0), so the simplest and most readable model was kept.

- **What drives the predictions.** Starting position first, then recent team and driver form. No experiment beat the chosen setup beyond the noise of 39 validation races.

Data: Kaggle / Ergast Formula 1 World Championship, 14 tables, 1950–2024 (1,125 races, 26,759 result rows). After cleaning, the analysis base is 24,586 race starts (Indianapolis 500 and drivers who did not start excluded).

1. Problem setup

Item	Decision
Unit of analysis	One driver entering one race; (raceId, driverId) is unique in the modeling table
Target	podium = 1 if results.positionOrder ≤ 3, else 0. positionOrder is the only position column filled for every entry
Prediction cutoff	After qualifying and the final grid, before the race start
Not allowed as features	Anything from the race being predicted: finish position, points, laps, times, status, pit stops, lap times, and post-race standings
Standings	Used only lagged: race *r* gets the standings published after the previous round of the same season (prev_raceId)
Time-based split	Train ≤ 2019, validation 2020–2021, test ≥ 2022
Base rate	12.7% of raw result rows are podiums. Modern races are near 15% (3 of ~20 cars); 1950s–80s fields with 30+ entrants are far lower

The split is by season, not random. A random split would put races from the same season (with the same cars and the same championship order) in both train and test, so the test score would overstate how well the model predicts a future season. Thresholds and hyperparameters are chosen on validation only.

2. Data audit and cleaning

All 14 tables were audited and cleaned on copies. A hash of every raw CSV, taken at load time and re-checked at the end, proves the raw files were never modified. 13 cleaned tables are written to data/cleaned/. seasons is dropped as fully redundant with races.year.

2.1 What the audit checked and fixed

Check	Result
Sentinels (\N)	160,076 literal \N cells converted to real missing values across 6 tables; 0 remain. qualifying also had 68 empty cells, treated the

Check	Result
	same way
Semantic typing	Dates → datetime. Race, lap, qualifying and pit times → numeric seconds. A strict converter fails if any non-missing value would be lost
Grain of `results`	85 (raceId, driverId) pairs had 2 rows (shared drives, 1950–1978). 91 rows removed: 26,759 → 26,668
Primary keys	Unique and non-null in all 13 cleaned tables, including composite keys
Foreign keys	17 links checked, including composite (raceId, driverId) links from qualifying, sprints, laps and pit stops to `results`; 0 orphans
Label fixes	"United States" → "USA" (circuits); "Argentinian " → "Argentine" (drivers)
Data errors fixed	2024 Belgian GP: 18 race times were stored as gaps to the disqualified winner, so his total time was added back. 2024 Monaco GP: `stop` and `lap` swapped in 7 pit-stop rows, proven from lap times. Two other races had pit-stop numbering that started at 2
Flagged, not deleted	Red-flag pit stops over 60 s, slow laps, McLaren's 2007 exclusion (`is_disqualified`)

EDA plots (cleaning). The figures below document the cleaning impact and a core exploratory finding from the results table; both come from the notebook audit / results EDA.

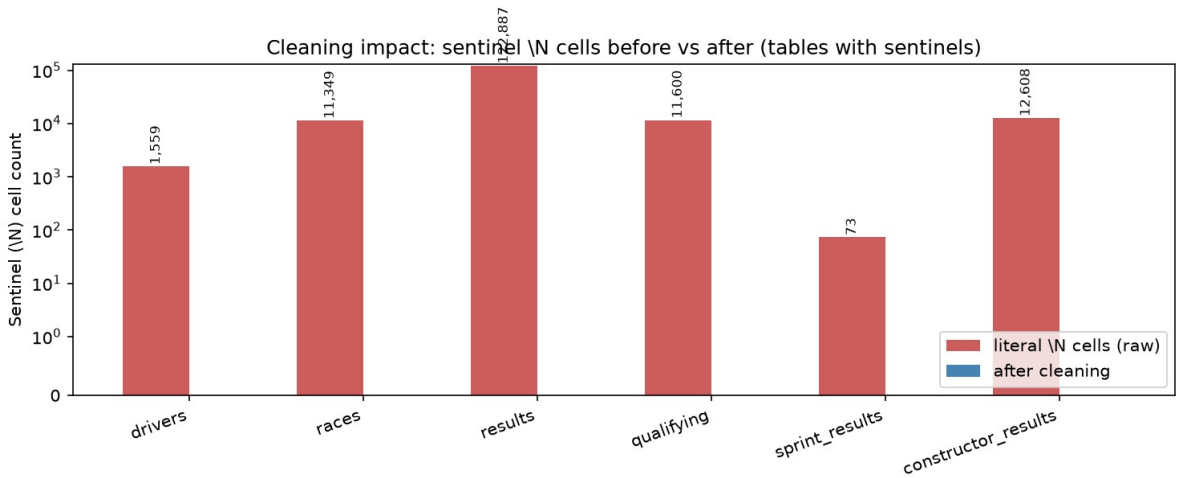


Figure A1. Cleaning impact: count of literal \N sentinel cells before cleaning versus after (zero remain). Notebook integrity audit §2.4.

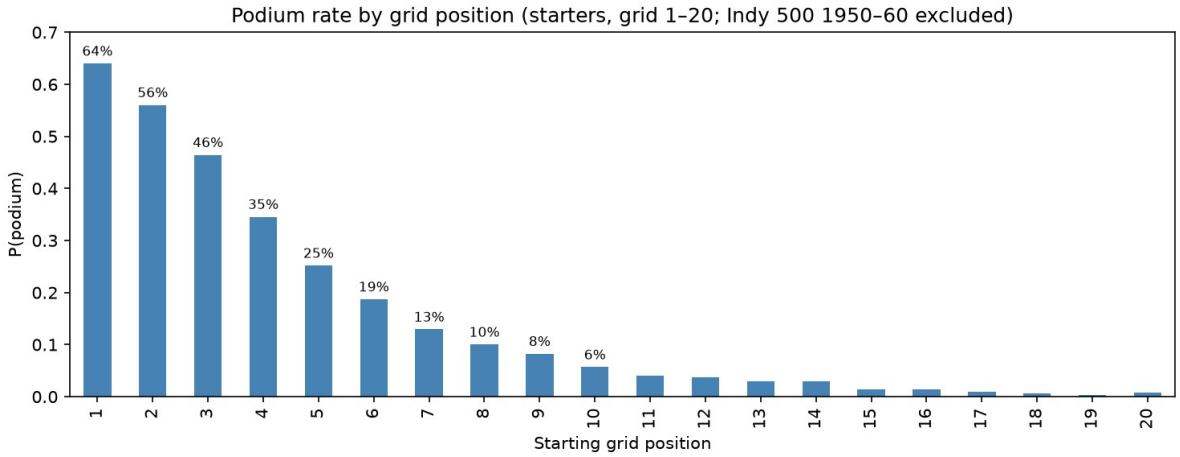


Figure A2. Podium rate by starting grid position among starters (grid 1–20; Indianapolis 500 1950–60 excluded). Pole converts much more often than the midfield — the main pre-race signal used later as a feature (notebook §1.7).

2.2 Shared-drive rule

Until 1957 a driver could take over a teammate's car mid-race, which gives one driver two result rows in one race. We keep one row per driver and race, in this order: the **best** `positionOrder` (the best result the driver achieved, which is exactly what the podium label asks about), then the row with **more laps**, then the **lowest** `resultId` as a deterministic tie-break. The brief's alternative, the car the driver started in, does not decide most pairs, because both rows usually have a grid slot or both have none. After deduplication, 18 early races still have 4–5 podium rows. This is correct: two drivers who shared one car both get the same `positionOrder` and both earned that podium.

2.3 Exclusions from the modeling table

- **Indianapolis 500 (1950–1960, 11 races):** flagged by race name, not by circuit, because the same track also hosted the regular US GP in 2000–2007. Almost no F1 regulars entered it, and its grid structure is different.
- **Drivers who did not start (1,701 rows):** `positionText` F (failed to qualify) or W (withdrew). They have no grid slot, so they cannot be predicted at the pre-race cutoff. The 87 pit-lane starters (`grid = 0` but did start) are kept and flagged.

2.4 Missingness is structural, not random

- `results.position` is missing for 41% of rows, exactly the unclassified drivers. This is why the target uses `positionOrder`.
- Race times exist only for lead-lap finishers. Fastest-lap data exists only from 2004.
- **Qualifying coverage drift:** `qualifying` starts in 1994, covers only 6–59% of starters in 1996–2002, and is complete from 2003. Only `grid` covers every year.
- Constructor standings start in 1958, the first Constructors' Championship. Practice, qualifying and sprint session dates exist only from 2021 and were dropped.

2.5 Leakage evidence from the EDA

The raw standings table is post-race: in round 1, standings points equal that race's points for 99.5% of rows. Lagging to the previous round removes the leak. The leaked version looks much stronger: the championship leader podiums 75% of the time on post-race standings vs 61% on pre-race standings. The sprint result is technically pre-race and strongly linked to the Sunday podium (61% vs 6.9%). It is left out because the training years (≤ 2019) contain no sprint weekends.

3. DEQ1: Front-row conversion by circuit, before and after 2014

Answer. From 2014 on, Monaco, Suzuka and Yas Marina convert a front-row start into a podium most often, and the Hungaroring least often. The ranking changed a lot after 2014: among the 14 circuits ranked in both periods, the 1950–2013 and 2014–2024 rates are uncorrelated (Spearman ρ = −0.11, p = 0.72).

Method. Join path: `results` → `races` (year, circuit) → `circuits` (name), plus `qualifying` for the qualifying view.

- **Front row by grid:** `grid` ∈ {1, 2}, the slot actually started from, after penalties. This is the main ranking, because grid exists for every year.
- **Front row by qualifying:** `qualifying.position` ∈ {1, 2}, the slot earned before penalties. It is used only for 2003–2024, where qualifying coverage is complete.
- **Conversion rate:** podiums divided by front-row starters, with Wilson 95% intervals.
- **Minimum sample:** a circuit is ranked in a period only with at least 7 races (14 front-row starts). Eleven seasons give at most ~22 starts per circuit after 2014, so a stricter cut would leave almost nothing to rank.

Period	Highest conversion	Lowest conversion	All circuits
2014–2024 (19 circuits ranked)	Monaco 89.5% (17/19), Suzuka 88.9% (16/18), Yas Marina 86.4% (19/22), Barcelona and Spa 81.8%	Hungaroring 59.1% (13/22), Mexico City 61.1% (11/18), Baku 62.5% (10/16)	76.2% of 453 starts
1950–2013 (37 circuits ranked)	Magny-Cours 82.9% (29/35), Istanbul Park and Indianapolis (US GP 2000–07) 78.6%, Suzuka and Sepang 70.0%	Detroit 28.6% (4/14), Jacarepaguá 35.0% (7/20), Brands Hatch 39.3% (11/28)	56.5% of 1,760 starts

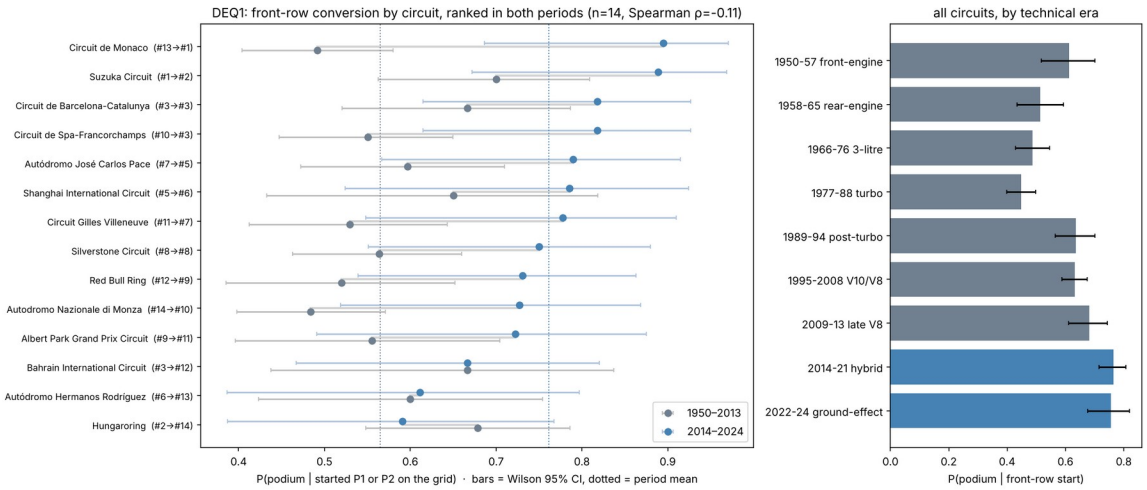


Figure 1. Left: the 14 circuits ranked in both periods; labels show each circuit's rank before → after 2014 within this common set. Right: front-row conversion across all circuits by technical era.

How the ranking changed. Monaco rose from #13 to #1 (49.2% → 89.5%, non-overlapping intervals), Spa from #10 to #3, and Canada from #11 to #7. The Hungaroring fell from #2 to #14 and Bahrain from #3 to #12. Suzuka and Barcelona stayed near the top in both periods.

Part of the shift is a level change across the whole field. Conversion was 45% in the 1977–88 turbo era and is about 76% in both the hybrid and ground-effect eras, as more reliable cars and dominant teams made a front-row start safer everywhere. For that reason the circuits are ranked within each period. A modern-only check (2003–2013 vs 2014–2024) still shows only weak persistence: ρ = 0.38 (p = 0.23) across 12 circuits. Monaco was already at 77% in 2003–2013, so part of its jump against the full 1950–2013 window comes from the low-reliability early decades.

3.1 Qualifying position vs grid position (2003–2024)

Of 856 drivers who qualified P1 or P2, **41 (4.8%) started further back** (median grid slot 7) after a penalty or a pit-lane start, and only **26.8%** of them reached the podium. The 36 drivers **promoted** onto the front row from P3 or lower converted at **63.9%**, close to the **72.9%** of drivers who both qualified and started there. The starting slot, not the qualifying result, carries the podium signal.

At circuit level the two definitions rank circuits almost identically: ρ = 0.91 in 2003–2013 and 0.88 in 2014–2024, with a mean difference of 2.3 percentage points. The largest gap is Interlagos 2014–2024 (13.9 points). Penalties rarely hit the front row, so they change individual outcomes much more than circuit rankings.

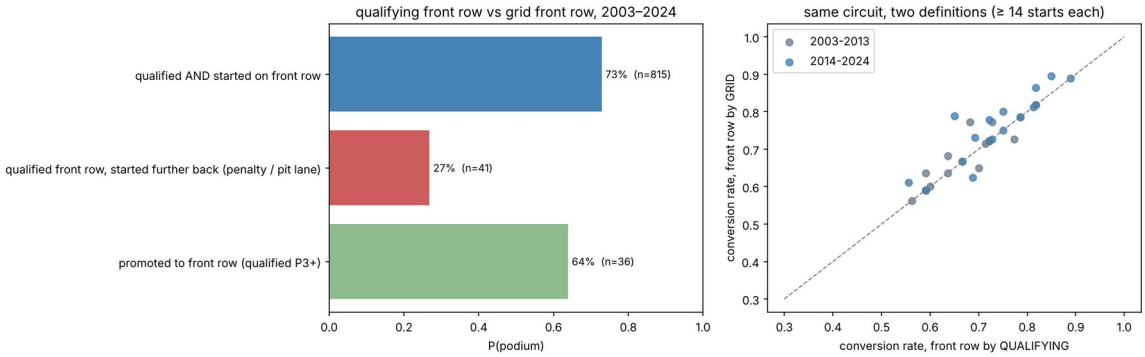


Figure 2. Left: podium rate for front-row qualifiers who kept, lost or gained a front-row slot. Right: each circuit's conversion rate under the two definitions; the dashed line marks equality.

3.2 Grain, missingness and cutoff

- **Grain:** one row per front-row starter per race. 1,089 races have exactly 2. 18 have only 1: in 5 the other front-row driver did not start, and in 13 no second car is recorded with grid 1–2 (for example, a pit-lane start recorded as `grid = 0`). 5 races from 1952–1964 have 3–4, because early records repeat grid numbers.
- **Missingness:** grid is complete for every year. The qualifying view is restricted to 2003+.
- **Cutoff:** grid and qualifying are both fixed before the start. The podium is the outcome being counted.
- **Limitations:** post-2014 samples are small (14–26 starts per circuit), so the intervals are about ±0.15–0.25 wide. Only large moves, such as Monaco up and the Hungaroring down, are clearly beyond noise. Circuit layouts also changed under the same `circuitId`.

4. DEQ2: Home-race podiums, controlling for starting position

Answer. No. Drivers do not podium more often at home once grid position is held fixed. The pooled odds ratio for home vs away within the same grid slot is **0.97 [0.81, 1.15]** (p = 0.72), and every controlled model gives an interval that contains 1.

Method. Join path: `results` → `races` → `circuits.country`, and `results` → `drivers.home_countries`. `home_countries` comes from the hand-built nationality → country map in cleaning; dual nationalities count both countries. A start is **home** when the circuit's country is one of the driver's home countries.

- **Base:** 24,499 starts, Indianapolis 500 excluded. The 87 pit-lane starts are also left out, because they have no comparable grid slot. 1,928 starts (7.9%) are at home.
- **Why a control is needed:** home drivers start further back (median grid 13 vs 11). In the big-field decades, home races drew many local one-off entrants who started near the back. A raw comparison therefore mixes "home" with "slow car".
- **Three controls:** (1) home and away rates inside the same grid band; (2) a Cochran–Mantel–Haenszel (CMH) odds ratio pooled across 21 grid-slot strata, which compares home and away drivers within the same slot and then combines the slots; (3) logistic regression `podium ~ home + C(grid)`, with standard errors clustered by driver, plus versions that also control for era and the team's pre-race championship position.

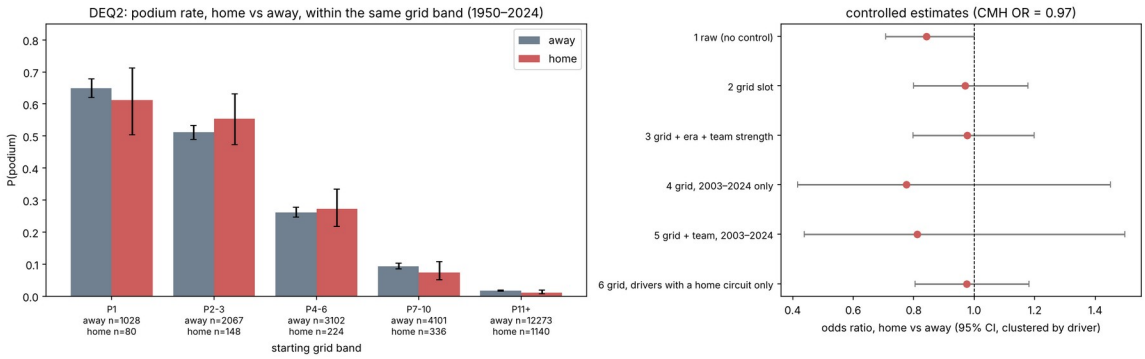


Figure 3. Left: podium rate inside each grid band, with Wilson 95% intervals. Right: odds ratio for home vs away under six specifications; 1 = no difference.

Specification	Starts	Odds ratio, home vs away [95% CI]	p
Raw, no control	24,499	0.84 [0.71, 1.00]	0.051
Grid slot	24,499	0.97 [0.80, 1.18]	0.75
Grid + era + team strength (lagged)	24,499	0.98 [0.80, 1.20]	0.82
Grid, 2003–2024 only	8,837	0.78 [0.42, 1.45]	0.43
Grid + team strength, 2003–2024	8,837	0.81 [0.44, 1.50]	0.50
Grid, only drivers whose country hosted a race	22,175	0.98 [0.81, 1.18]	0.80

Reading the result. The raw comparison makes home drivers look *worse* (11.9% vs 13.9%). That gap disappears once grid is controlled for, so it was caused by where home drivers start, not by being at home. Within the grid bands there is no consistent direction: home is slightly higher in P2–3 and P4–6, slightly lower in P1, P7–10 and P11+, and every pair of intervals overlaps. A Breslow–Day test (p = 0.16) finds no evidence that the home effect differs by grid slot. The data rule out a home advantage larger than roughly +15–20% in the odds of a podium at the same starting position.

4.1 Grain, missingness and cutoff

- **Grain:** one row per (raceId, driverId) start.
- **Missingness:** 50 drivers come from nationalities with no F1 circuit (e.g. Finnish, New Zealander) and are always "away". Excluding them changes nothing (last row of the table). Lagged team standings are missing for season openers and pre-1958 races; these are modeled as their own category, not dropped.
- **Cutoff:** nationality, circuit, grid slot and lagged standings are all known before the start.
- **Limitations:** nationality is not the same as where a driver lives or where their team is based. The 2003+ estimate rests on only 363 home starts. This is an association, not a causal estimate: a null result means no detectable home effect at the same grid slot, not proof that a crowd has no influence.

5. DEQ3: Mechanical-retirement rate by constructor, 2014–2021 vs 2022–2024

Answer. In 2014–2021 the highest mechanical-retirement rates were **Lotus F1 (20.0%), Toro Rosso (16.1%) and Renault (14.1%)**. In 2022–2024 they were **Alfa Romeo (12.5%), Alpine (10.4%) and AlphaTauri (8.0%)**. Reliability improved across the field, from 9.6% to 5.9% of starts (z = 4.06, p < 0.001).

Method. Join path: `results` → `status.status_group` (the hand taxonomy from cleaning) → `constructors` (team name), with `races.year` giving the period.

- **Mechanical retirement:** a race start that ended because a car component failed (Engine, Power Unit, Gearbox, Hydraulics, Brakes, …). Crashes, driver errors and regulation decisions are excluded. For example, `Puncture` and `Collision damage` count as incidents, and `Disqualified` and `Withdrew` count as other.
- **Rate:** mechanical retirements divided by race starts, per constructor and period. Drivers who did not start are excluded from the denominator. Sprint races are not included, since `results` holds the Grand Prix only.
- **Minimum sample:** 40 starts in the period, roughly one full season for two cars. Caterham (7/33) and Marussia (3/31), both 2014 only, are below it and are not ranked.
- **Team identity** follows `constructorId`, as decided in cleaning, so a renamed team is a separate row. A second view groups each team on the 2024 grid with its earlier names.

Period	Highest rate	Lowest rate	Field
2014–2021	Lotus F1 20.0% (15/75), Toro Rosso 16.1% (39/242), Renault 14.1% (28/199), McLaren 13.6% (43/316)	Aston Martin 2.3% (1/44, 2021 only), Mercedes 4.1% (13/320), Alfa Romeo 4.2% (5/120), Ferrari 6.3% (20/318)	9.6% (311 / 3,246)
2022–2024	Alfa Romeo 12.5% (11/88), Alpine 10.4% (14/135), AlphaTauri 8.0% (7/88)	Sauber 2.1% (1/48, 2024 only), McLaren and Red Bull 3.7% (5/136 each), RB 4.2% (2/48)	5.9% (80 / 1,352)

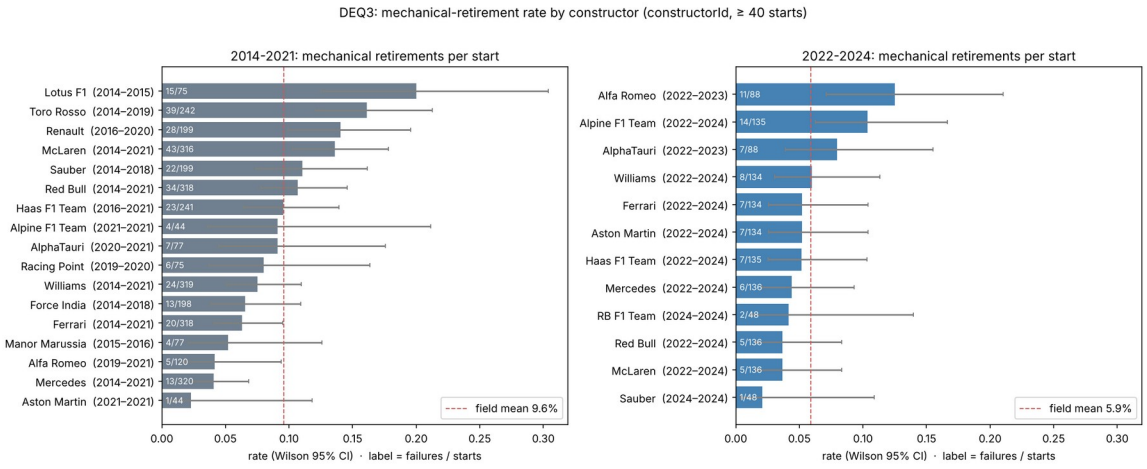


Figure 4. Mechanical retirements per start, with Wilson 95% intervals; labels give failures / starts. Constructors with at least 40 starts in the period.

5.1 The same question by today's teams

Grouped by current team, the ranking only weakly carries over between periods (ρ = 0.24). **Enstone (Lotus F1 → Renault → Alpine) is the least reliable lineage in both periods** (14.8% → 10.4%). The largest improvements are McLaren (13.6% → 3.7%), Faenza (Toro Rosso / AlphaTauri / RB, 14.4% → 6.6%) and Red Bull (10.7% → 3.7%). Mercedes is among the most reliable in both periods (4.1% → 4.4%). The failure mix also shifted: `Engine` grew from 18.6% to 26.2% of mechanical retirements and `Brakes` fell from 14.1% to 5.0%, while `Hydraulics` and `Undertray` (the ground-effect floor) appear more often after 2022.

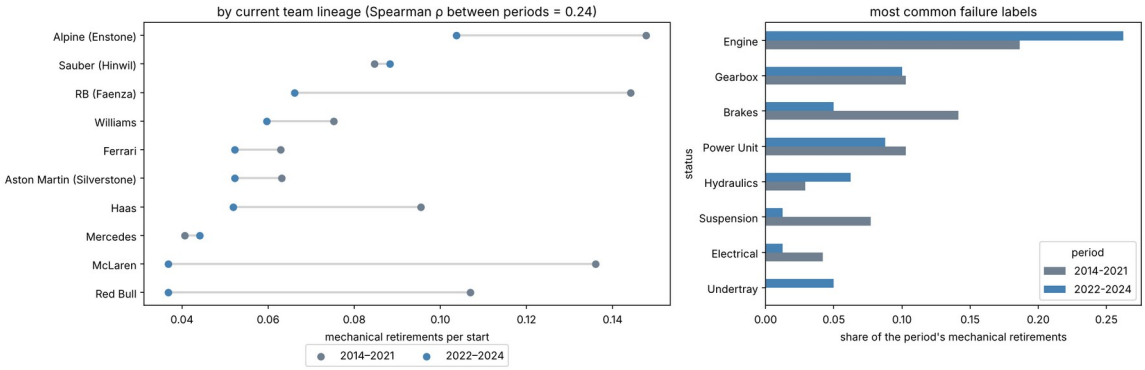


Figure 5. Left: each 2024 team grouped with its earlier names, 2014–2021 vs 2022–2024. Right: share of each period's mechanical retirements by failure label.

5.2 Grain, missingness and cutoff

- **Grain:** one row per `(raceId, driverId)` race start, so each car counts once per Grand Prix.
- **Retirements with no cause recorded:** 49 rows in 2014–2024 carry only the label `Retired` (37 in 2014–2021 and 12 in 2022–2024, 9 of them in 2024). They are excluded, so the rates above are lower bounds. Counting all of them as mechanical leaves the 2014–2021 ranking intact (ρ = 0.97) but reshuffles the lower 2022–2024 places (ρ = 0.66): RB enters the top three. Alfa Romeo and Alpine stay at the top under both rules.
- **Cutoff:** status is post-race. Here it is the outcome being counted, and it is never used as a model feature.
- **Limitations:** 2022–2024 counts are small (1–14 failures per team), so most intervals overlap. Only Alfa Romeo and Alpine sit clearly above the most reliable teams. The mechanical/non-mechanical split depends on the hand taxonomy and on how each retirement was labelled in the source data.

6. Feature engineering

The modeling table has **one row per driver who started a race** (24,563 rows, unique on (raceId, driverId)) and **27 numeric features in 8 groups**. The groups are the units of the ablation in section 9. Every feature follows three rules: (1) it is known after qualifying and before the start; (2) anything built from results uses **earlier races only** (`shift(1)` before averaging; standings lagged to the previous round); (3) every missing value has a documented meaning (6.2). An assert checks that no post-race column is in the table.

6.1 Final feature set

Evidence is measured on the **training seasons only** (≤ 2019), so validation and test never influence feature choices. ρ is the Spearman correlation with podium on train.

Group	Features	What they measure	Evidence (train)	Why valid at the cutoff
Grid	grid_rank grid_pct field_size front_row is_pit_start	Starting slot among the actual starters (pit-lane starters at the back), slot as a share of the field, field size	Strongest single signal: ρ = −0.46. Front row 58.9% podium rate vs 9.0% for the rest; fields of ≤ 16 cars 20.6% vs 9.7% for 29+	The grid, including penalties, is published before the start
Qualifying	quali_gap_pct grid_minus_quali has_quali	Gap to pole in % (capped at 7%), places lost or gained between qualifying and grid, flag for a qualifying record	Gap ρ = −0.47. Within the same grid band, closest-to-pole drivers podium far more (P4–6: 29.8% vs 17.4%). In the midfield and back, penalised drivers podium 2–3× more than drivers who qualified in the same slot	Qualifying ends before the race. Qualifying *position* is left out: it is nearly identical to grid
Standings	drv_points_share con_points_share drv_champ_pos con_champ_pos	Points as a share of the championship leader, and championship position, for driver and team	Driver ρ = +0.36 (share) and −0.40 (position). Championship leader podiums 58.9%. Shares make 1950 (8 points for a win) and 2023 (25) comparable	Standings after the **previous round** only; season openers use last season's final position
Form	drv_podium_ewm drv_finish_frac_ewm con_podium_ewm con_finish_frac_ewm	Recent podium rate and finishing share of the driver and the team (exponential average, half-life 5 races)	Driver ρ = +0.42, team +0.40. The EWM beat every fixed window as a single feature (10 definitions compared). Adds signal inside every grid band (P4–6: 39.6% high form vs 11.4% low)	shift(1) before averaging; verified by rebuilding 10 races with their own results scrambled

Group	Features	What they measure	Evidence (train)	Why valid at the cutoff
Reliability	con_mech_dnf_ewm drv_incident_ewm	Team's recent mechanical-failure rate (DEQ3 taxonomy) and driver's recent crash/incident rate	Team ρ = −0.15, driver −0.07; weaker but a different signal from pace	Status of **earlier** races only; same rebuild check as form
Circuit history	drv_circuit_visits drv_circuit_last_finish drv_circuit_avg_finish	Earlier starts at this circuit and finishing share at the last / all earlier visits	ρ = −0.20 / −0.22 on their own, but little beyond grid and form	Earlier visits only (shift(1) per driver and circuit)
Driver	drv_starts_before age teammate_grid_gap has_teammate	Experience, age, grid gap to the teammate (same car), single-car flag	Experience ρ = +0.19 (first start 2.7% podium rate, 100+ starts 22.2%); teammate gap separates strong cars within a grid band	Career history, date of birth and the published grid
Context	is_home_race season_progress	Home race (DEQ2 map) and round / rounds in the season	Weak on their own (|ρ| ≤ 0.02). Standings become more predictive later in a season (ρ 0.24 → 0.44)	Nationality and calendar are known in advance

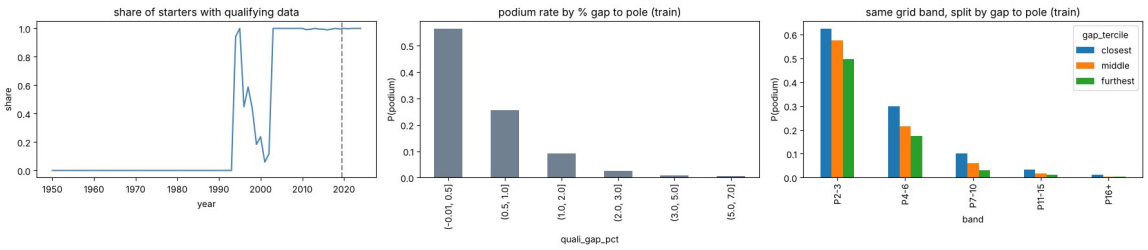


Figure 6. Qualifying (training seasons). Left: share of starters with qualifying data per season; the dashed line marks the end of training. Middle: podium rate by gap to pole. Right: within the same grid band, drivers closest to pole podium more often, so the gap adds pace information beyond the grid.

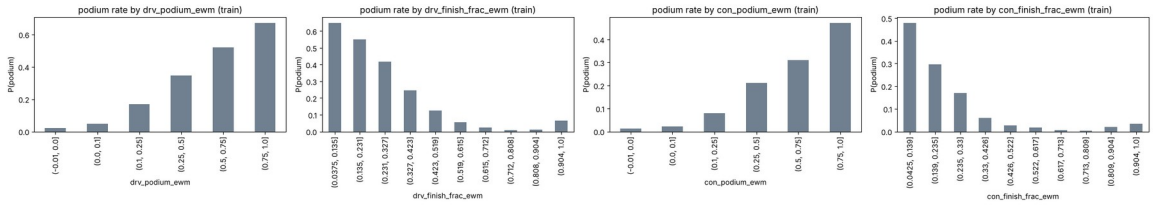


Figure 7. Form features (training seasons): podium rate rises steadily with recent driver and team podium form and falls with recent finishing share.

6.2 Missing values

Every gap has a known cause, so it is filled with **real pre-race information** wherever it exists, not a statistic:

- **Season openers:** championship position = last season's final position (last year's champion podiums 55.0% at the opener, unranked drivers 3.2%).
- **1950–57 constructors' standings:** rebuilt from race results, since there was no constructors' championship yet.
- **Debut form:** the teammate's form in the same race (same car), then team form, then the base rate (3 / field size).
- **First circuit visit:** the driver's general finishing form; **single-car entries:** teammate gap 0 plus `has_teammate = 0`.
- **Only the qualifying gap has no pre-race substitute.** It is missing for 63.7% of training rows (no qualifying data before 1994) but about 1% of validation and test. It is imputed inside the model pipeline with the **median gap of the driver's grid band**, learned on training rows only (section 7.1).

The same rebuild check (10 races, own results scrambled, later races removed) gives identical filled values, so the fills add no leakage.

6.3 Deliberately excluded

- **Post-race fields:** finish position, points, laps, times, status, fastest lap, pit stops, lap times, constructor results and unlagged standings. All leak the result.
- **Sprint result:** pre-race and predictive, but the training seasons contain no sprint weekends, so the model could not learn a weight for it.
- **Qualifying position:** almost identical to `grid_rank`; only the difference (`grid_minus_quali`) is kept.
- **Identifiers and era:** driverId, constructorId, circuitId and the era label. Teams are renamed, and the test seasons are an era that never appears in training, so an id- or era-level effect would not carry over. Era differences that matter are captured by field size, points shares and reliability rates.

6.4 Train / validation / test split

Split	Seasons	Races	Rows	Podium rate
Train	1950–2019	1,006	22,436	0.135

Split	Seasons	Races	Rows	Podium rate
Validation	2020–2021	39	775	0.151
Test	2022–2024	68	1,352	0.151

The split is by season (section 1). The test seasons are exactly the new ground-effect regulation era, so the test score also measures how the model copes with a rule change. Validation has only 39 races, so all choices made on it are read against a **race-level bootstrap**: whole races are resampled 500–1,000 times, and a difference counts only if its 95% interval excludes 0.

7. Preprocessing

7.1 Pipeline

Everything is one scikit-learn `Pipeline`, **fitted on the training rows only** and applied unchanged to validation, test and the inference function:

- **Grid-band imputer** for `quali_gap_pct`: the median gap per grid band (P1 0.00%, P7–10 1.58%, P16+ 4.11%). One global median (1.96%) would tell a pole-sitter and a back-marker they had the same pace and weakens the feature's correlation with podium from ρ = −0.47 to −0.24; the grid-band fill keeps it at −0.44.
- **Median imputer** as a safety net for any other missing value (needed for partially-missing records at inference).
- **For logistic regression and the neural network only:** `log1p` on the two count features (skewness of `drv_starts_before` 1.42 → −0.74) and standard scaling. Tree models split on value order, so they only get the imputation.

Scaling on train also makes the era shift visible: in the scaled units, validation and test cars are far more reliable (`con_mech_dnf_ewm` mean −1.17 / −1.25) and drivers more experienced (+0.65 / +0.86).

7.2 Preprocessing-order experiments

A plain logistic regression is used as a probe, with everything else fixed; results on validation:

Experiment	Variant	Validation PR-AUC	Δ vs used [95% CI]
1. Which rows the preprocessing is fitted on	Training rows only (used)	0.7276	—
	All rows before splitting (leaky)	0.7272	[−0.002, +0.001]
2. Order of the two imputers	Grid band, then median (used)	0.7276	—

Experiment	Variant	Validation PR-AUC	Δ vs used [95% CI]
	Median, then grid band	0.7286	[−0.005, +0.007]

Fitting on all rows changes what is learned (the gap imputed for P16+ drops from 4.11% to 3.84%) but barely moves the score, because training is 91% of the data. It is still the only valid procedure: leaky statistics let the held-out seasons into the model. Imputer order changes the data (with the median first, every gap gets 1.96%) but not the score significantly, because validation and test have real gaps for about 99% of rows. Grid band first is kept because it gives sensible values where they matter: at inference and in the explanations.

8. Models and evaluation

8.1 Metrics

Primary: PR-AUC. Podiums are only 13.5–15.1% of rows. PR-AUC measures how well real podium finishers are ranked above everyone else, and a random model scores the base rate (0.151), not 0.5. All hyperparameters are chosen on it. **Secondary: F1** at the threshold that maximises F1 on validation, applied unchanged to test, because the inference function must output yes/no. ROC-AUC, precision and recall are reported for every model; ROC-AUC is above 0.9 for every model, including the grid rule, so it cannot separate them.

8.2 Models

Role	Model	How it works	Tuned on validation (chosen)	Limitations
Reference	Grid rule	P(podium) = podium rate of the starting slot in training	None	Uses nothing but the grid
Baseline	Logistic regression, default (C = 1), all 27 features	Weighted sum of the scaled features turned into a probability by the logistic function	None	Additive and one-directional: no interactions (e.g. penalty × grid slot); correlated features split their weight
Statistical 1	Logistic regression, tuned	As above, with L2 strength C	C ∈ {0.01, 0.1, 1, 10} × class_weight {none, balanced} → C = 10, no weighting	As above
Statistical 2	Random forest (300 trees)	Average vote of trees, each on a bootstrap sample	min_samples_leaf {5, 20, 50, 100, 200} × max_features {sqrt,	Small leaves memorise (train PR-AUC 0.93–0.96);

Role	Model	How it works	Tuned on validation (chosen)	Limitations
		and random feature subsets	0.5} → 200, sqrt	probabilities are averaged votes; cannot extrapolate
Statistical 3	Gradient boosting (HistGradientBoosting)	Trees built in sequence, each correcting the errors so far (learning rate 0.05, L2 1.0)	max_leaf_nodes {7, 15, 31} × min_samples_leaf {20, 50, 200} × trees {100, 300} → 7, 20, 300	Most flexible, easiest to overfit (31-leaf trees reach train PR-AUC up to 0.87)
Shallow NN	MLP, 1–2 hidden layers (ReLU, Adam)	Feed-forward network on the scaled features; early stopping on 10% of the training rows	(32) vs (64, 32) × L2 alpha {1e-4, 1e-2} → (32), alpha 1e-4	Needs scaling; result depends on the random seed; early stopping monitors accuracy; not directly interpretable

Every model is trained on the **same rows and the same 27 features**. Each tuning grid and its train/validation scores are kept in the notebook, so every before/after comparison is visible.

8.3 Results

Model	Split	ROC-AUC	PR-AUC	F1	Precision	Recall
Grid rule (reference)	Train	0.884	0.500	0.544	0.546	0.543
	Validation	0.912	0.649	0.701	0.701	0.701
	Test	0.914	0.647	0.652	0.652	0.652
Logistic regression (baseline)	Train	0.903	0.574	0.590	0.493	0.735
	Validation	0.930	0.728	0.719	0.662	0.786
	Test	0.937	0.711	0.665	0.545	0.853
Logistic regression (tuned), final	Train	0.903	0.574	0.590	0.492	0.736
	Validation	0.930	**0.728**	0.716	0.657	0.786
	Test	0.937	**0.711**	0.667	0.545	0.858
Random forest	Train	0.911	0.600	0.592	0.488	0.752

Model	Split	ROC-AUC	PR-AUC	F1	Precision	Recall
	Validation	0.933	0.726	0.715	0.644	0.803
	Test	0.939	0.723	0.668	0.536	0.887
Gradient boosting	Train	0.925	0.652	0.606	0.478	0.829
	Validation	0.934	0.720	0.714	0.638	0.812
	Test	0.939	0.717	0.636	0.512	0.838
Shallow neural network	Train	0.903	0.574	0.584	0.512	0.679
	Validation	0.923	0.720	0.708	0.683	0.735
	Test	0.932	0.695	0.656	0.616	0.701

Thresholds chosen on validation: 0.46 (grid rule), 0.25 (both logistic regressions, random forest), 0.21 (gradient boosting), 0.34 (neural network). The test set was scored once per model and never used for a choice.

Model	Validation Δ PR-AUC vs baseline [95% CI]	Test Δ PR-AUC vs baseline [95% CI]
Grid rule	−0.078 [−0.130, −0.024]	−0.064 [−0.110, −0.017]
Random forest	−0.002 [−0.018, +0.018]	+0.013 [−0.006, +0.032]
Gradient boosting	−0.007 [−0.035, +0.025]	+0.006 [−0.020, +0.030]
Shallow neural network	−0.008 [−0.051, +0.029]	−0.015 [−0.049, +0.015]

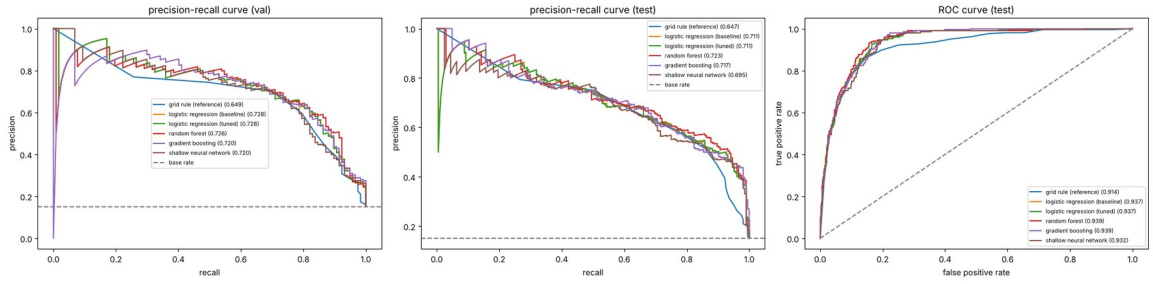


Figure 8. Precision–recall curves on validation and test, and ROC curves on test, for every model (PR-AUC / ROC-AUC in the legend). The dashed line is the base rate.

Reading the results.

- **The features matter beyond the grid:** every learned model beats the grid rule clearly (+0.078 PR-AUC on validation, +0.064 on test for the baseline; both intervals exclude 0).
- **The learned models are tied:** every difference to the baseline has an interval containing 0. With 39 validation and 68 test races, the more flexible models do not earn a measurable gain over logistic regression.
- **Overfitting was controlled by tuning:** small-leaf forests and large-leaf boosting reached train PR-AUC 0.81–0.96 with no gain on validation; the chosen settings keep train and validation close.
- **Train scores below validation** (0.57 vs 0.73 for logistic regression) are not underfitting: the training seasons are harder (bigger fields, more retirements, 64% without qualifying data; section 8.5).

8.4 Final model

Logistic regression (tuned): highest validation PR-AUC (0.728, the selection criterion), the simplest model, directly readable weights, and well-calibrated probabilities (8.5). Random forest is statistically tied and is used as a second model in the experiments and explanations. On the test seasons the final model reaches **PR-AUC 0.711, ROC-AUC 0.937 and F1 0.667** (precision 0.545, recall 0.858).

8.5 Error analysis

- **Stable across seasons:** every held-out season scores far above its 0.15 base rate (0.709, 0.756 on validation; 0.702, 0.747, 0.675 on test). 2022, the first season of the new regulations, scores like the validation seasons. 2024, with four different race-winning teams, is the hardest.
- **Era (in-sample):** the model fits early eras far worse (PR-AUC 0.45–0.47 for 1958–1988) than the hybrid era (0.735). Before 1994 there is no qualifying data and retirements were frequent, so a podium was less predictable before the race.
- **Calibration:** all four models have Brier scores of 0.069–0.071 (no-skill 0.128). The final model's probabilities sum to 2.93 per race on average (truth: exactly 3), so they can be read as real chances; favourites are slightly under-predicted (predicted ≈ 0.67, observed ≈ 0.76).
- **Biggest misses:** of 146 test false alarms at the threshold, 65% finished P4–P6 and 16% retired. The most confident miss is Verstappen from pole in Australia 2024 (P = 0.85, mechanical failure). The biggest surprises are wet races (São Paulo 2024: Verstappen won from P17, Ocon and Gasly on the podium; Zandvoort 2023) and grid penalties (Leclerc, Abu Dhabi 2024, P19 → P3).
- **Threshold robustness:** F1 stays within 0.02 of its best for thresholds 0.21–0.45 on validation and 0.24–0.41 on test; the chosen 0.25 lies in both. Test F1 at 0.25 (0.667) is within 0.003 of the best any threshold could reach on test (0.670).

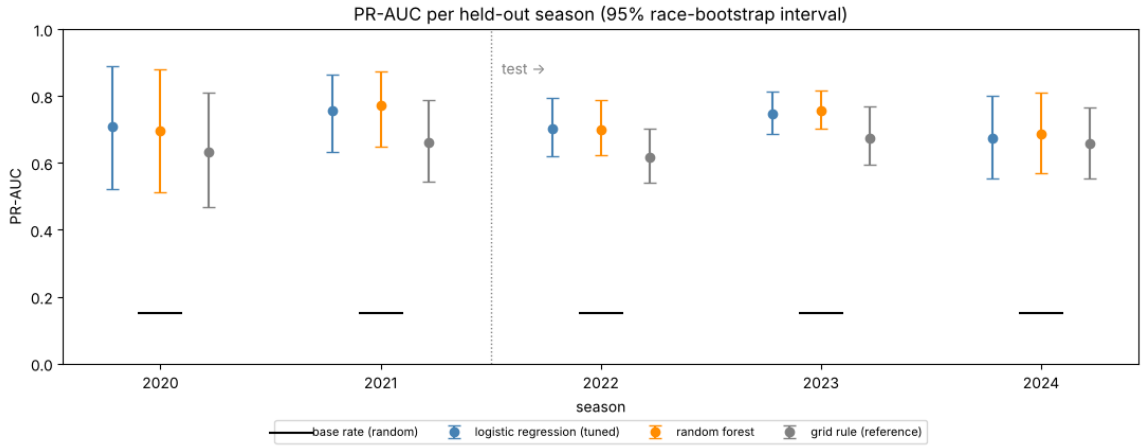


Figure 9. PR-AUC per held-out season with 95% race-bootstrap intervals; black marks are each season's base rate.

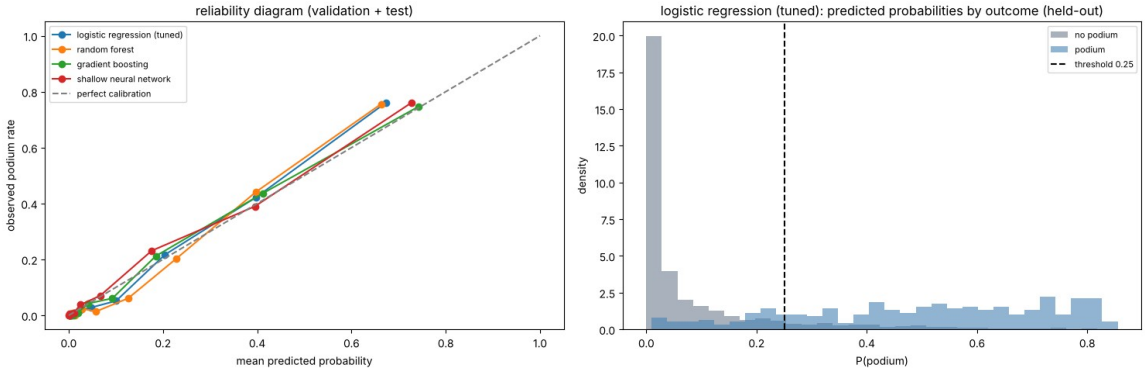


Figure 10. Left: reliability diagram on validation + test (perfect calibration on the diagonal). Right: predicted probabilities of the final model for podium and non-podium drivers, with the threshold.

9. Experiments

Each experiment changes one thing and keeps the rest fixed (same rows, tuned hyperparameters and pipeline). A variant replaces the current setup only if it raises validation PR-AUC **and** its 95% race-bootstrap interval excludes 0. Logistic regression (final) and random forest are both run; test is shown for information only.

9.1 Feature-group ablation

Removed	LR validation Δ	LR test Δ	RF validation Δ	RF test Δ
(none: full model)	0.728	0.711	0.726	0.723
Grid	−0.045	−0.044	−0.017	−0.079
Qualifying	−0.004	−0.002	−0.005	−0.000
Standings	+0.001	−0.002	+0.005	−0.001

Removed	LR validation Δ	LR test Δ	RF validation Δ	RF test Δ
Form	−0.004	−0.005	+0.010	+0.001
Reliability	−0.001	+0.002	+0.001	+0.003
Circuit history	−0.002	−0.001	−0.002	+0.004
Driver	−0.005	−0.001	+0.001	−0.001
Context	−0.003	−0.000	−0.003	+0.003
All except grid (grid only)	**−0.073** †	−0.064	**−0.075** †	−0.061

First row: full-model PR-AUC; other rows: change when the group is removed and the model retrained. † = 95% validation interval excludes 0.

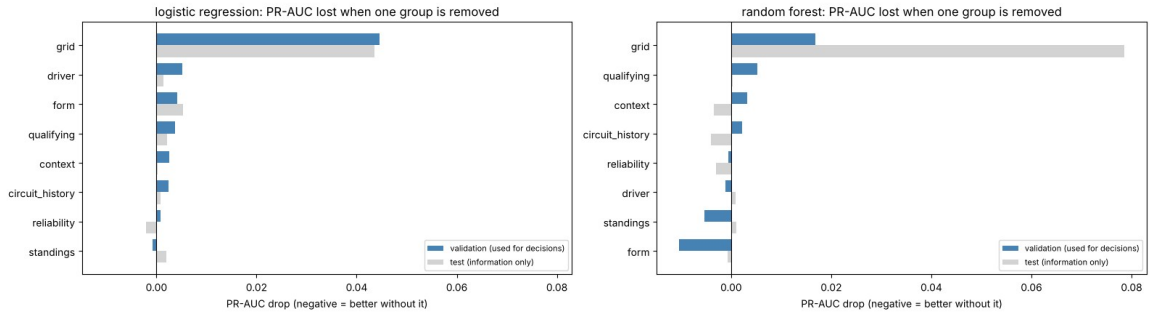


Figure 11. PR-AUC lost when one feature group is removed and the model retrained (blue: validation, used for decisions; grey: test).

- **Grid is the only group that matters on its own:** removing it costs about 0.045 PR-AUC for logistic regression on both splits.
- **Every other group alone changes PR-AUC by about 0.005 or less**, because they overlap: standings, form, qualifying gap and reliability all describe how strong the car and driver are (correlations of 0.6–0.9 between them).
- **Together they matter:** keeping only the grid features drops the model to about 0.65 (validation interval [−0.128, −0.013]). Grid gets a model to about 0.65; the other seven groups together add the rest. All 8 groups are kept, since no removal improves validation beyond noise.

9.2 Other experiments

Experiment	Result (logistic regression unless stated)	Decision
Qualifying handling: no qualifying features; train only on 2003–19 or 1994–19	All worse than the current setup on validation (−0.004, −0.017, −0.010) and test. Random forest on 2003+ gains +0.016 on validation (interval [+0.000, +0.029]) but loses 0.007 on test	Keep all seasons + grid-band imputation (all models must share the same rows)

Experiment	Result (logistic regression unless stated)	Decision
Penalty × grid interaction (`penalty_x_grid`, `quali_rank`)	No gain (−0.001 / −0.000). Penalised back-half starters: 3.7% podium rate, already predicted at 4.0%: form and standings already mark them as fast cars	Keep 27 features
Class weighting (balanced)	PR-AUC 0.728 → 0.725, but probabilities double (mean 0.147 → 0.310 vs real 0.151), Brier 0.068 → 0.111, test F1 0.667 → 0.647	No weighting; the threshold handles imbalance
Decision rule: top 3 per race instead of the threshold	Test precision 0.545 → 0.652, recall 0.858 → 0.652, F1 0.667 → 0.652 (threshold flags 4.7 drivers per race on test)	Keep the F1 threshold; top-3 remains an option for users

10. Explainability

Importance is not causation. Every method below describes how *the model* uses its inputs. A feature being important means the model relies on it to predict podiums, not that it *causes* podiums: grid position, form and standings are all consequences of having a fast car and a good driver.

10.1 Global explanations

- **Permutation importance** (validation, 20 repeats): shuffling `grid_rank` costs 0.156 PR-AUC; team finishing form, grid share and driver podium form 0.03–0.04 each; everything else under 0.01. Shuffling **whole groups** (needed because grid rank and grid share correlate at 0.97): grid 0.34, form 0.17, standings 0.02, others ≤ 0.01. Random forest ranks the groups in the same order.
- **Why form matters here but not in the ablation:** permutation scrambles a feature in an already-trained model, which cannot compensate; the ablation retrains, and the overlapping groups take over. The model *relies* on form, but form is *replaceable*.
- **SHAP** (exact `LinearExplainer` for logistic regression, `TreeExplainer` for random forest): the same top four as permutation importance (grid rank, team form, grid share, driver form). Directions are sensible: better grid and form push toward a podium, a pit-lane start strongly away.
- **A warning example:** reliability and has-qualifying look large in SHAP but not in permutation importance. SHAP compares with an average *training* row (from 1950), and every modern car is more reliable and has qualifying data, so these features shift *every* modern driver equally without changing who ranks first. The model even gives a higher team failure rate a slightly positive weight: given team finishing form (which counts breakdowns as bad finishes), a team whose poor results come from failures is faster than its form suggests. This is a statistical correction between correlated inputs, not a causal effect.

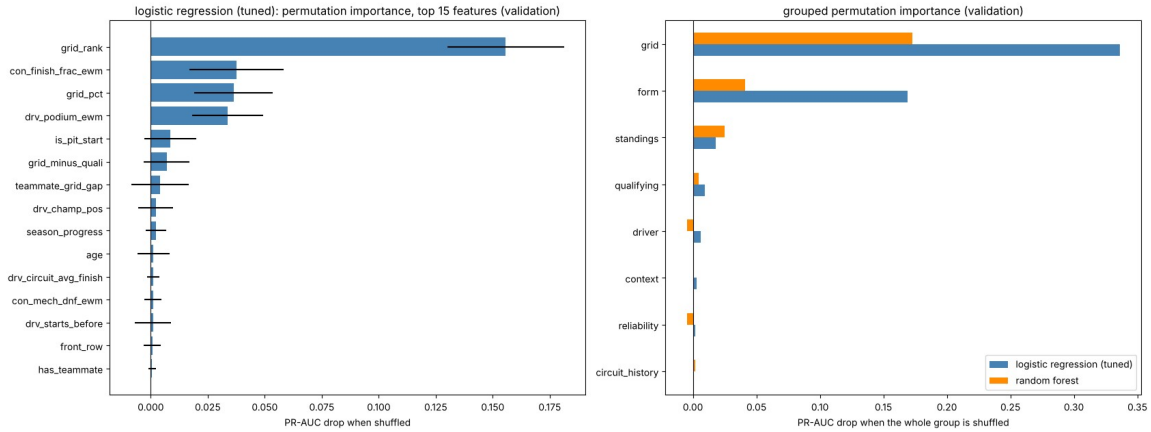


Figure 12. Left: permutation importance of single features (final model, validation). Right: grouped permutation importance for both models.

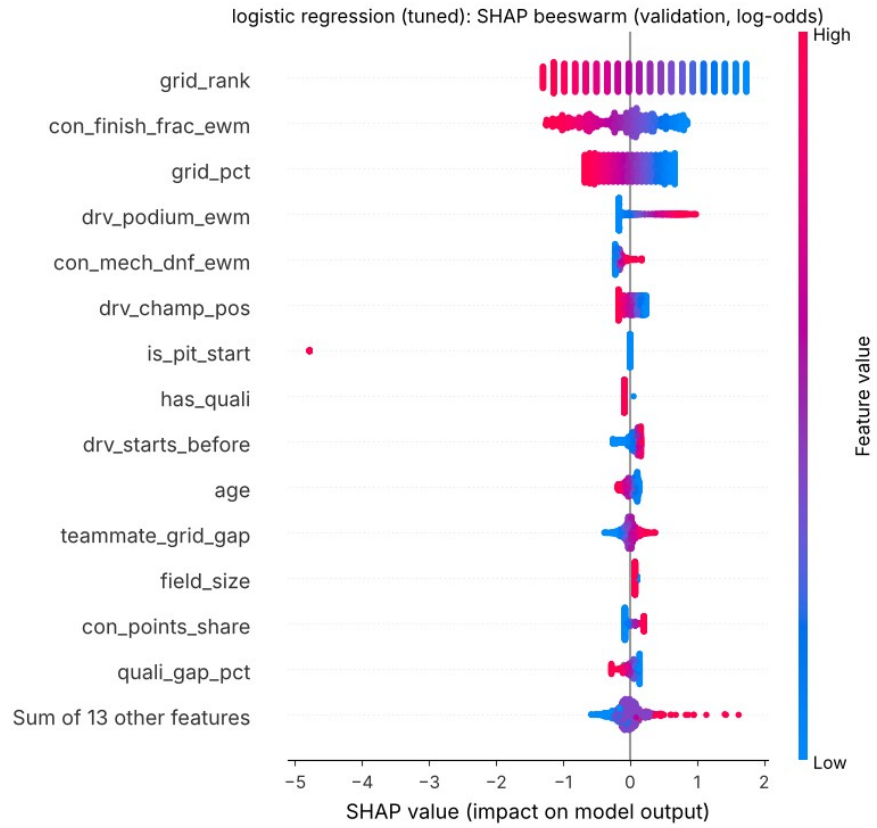


Figure 13. SHAP beeswarm for the final model on validation (log-odds; colour = feature value).

10.2 Local explanations

- **Case A, Verstappen, Belgium 2023:** fastest in qualifying, 5-place gearbox penalty, started P6, won; P(podium) = 0.72. SHAP: driver form +0.98, grid rank +0.94 (P6 is still well above average), team form +0.89, and `grid_minus_quali` = +5 adds +0.47: the model reads the penalty as a fast car out of position.

- **Case B, Verstappen, Australia 2024:** pole, retired with a mechanical failure; P(podium) = 0.85, almost all from pole (+1.74) and form. His team's recent failure rate was about 0.1%: nothing before the race pointed to the failure.
- **LIME:** with LIME's default binned sampling, its local surrogate reproduced the model badly (0.31 vs 0.72, local R^2 0.53). Sampling small changes around the driver (clipped to the training range) raised the fit to R^2 0.86 / 0.70, and its top four features then match SHAP's. SHAP answers *what each feature contributed compared to an average driver*; LIME answers *what would happen if the feature changed a little*, so their signs can differ (P6 is better than average, but starting further back would hurt).

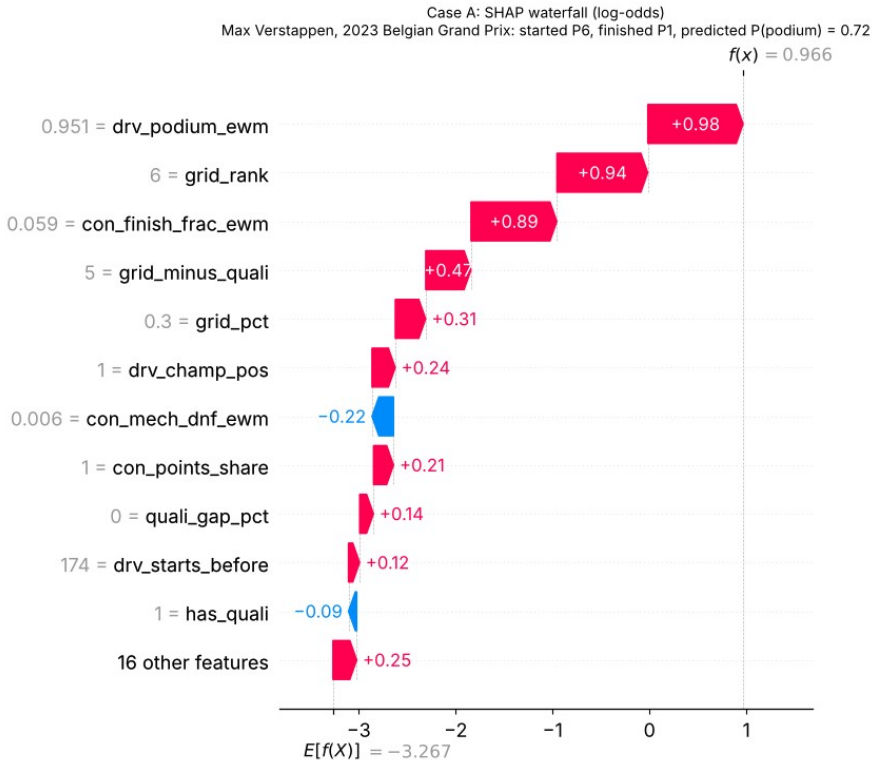


Figure 14. SHAP waterfall for case A (log-odds): from the average prediction to Verstappen's 0.72 at Spa 2023.

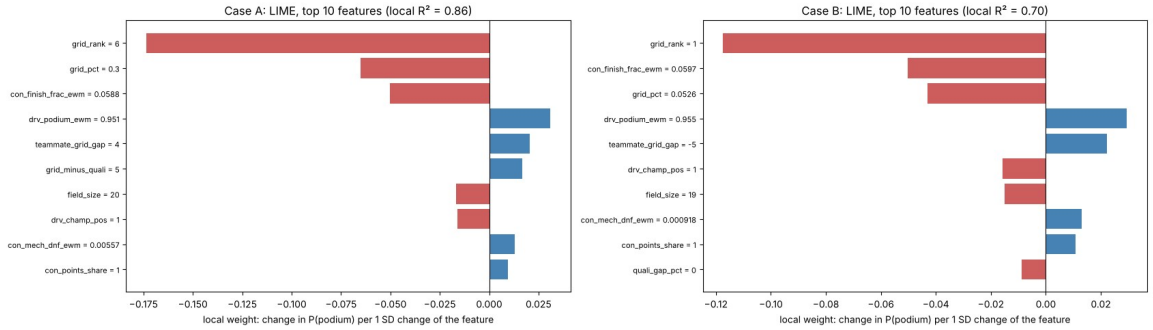


Figure 15. LIME for cases A and B: local weights of the top 10 features (effect on P(podium) per 1 SD change).

11. Inference function

`predict_podium(record)` takes one driver's pre-race information as a dictionary and returns the probability, the yes/no decision at the validation threshold (0.25), and the list of features that were **derived** or **imputed**. It rejects unknown field names and non-numeric values, derives `grid_pct` and `front_row` exactly from the grid, and fills everything else with the fitted pipeline (grid-band gap, training medians); nothing is refitted.

Demo	Input	Output	Note
Complete record	Verstappen, Belgium 2023, all 27 features	P = 0.724 → podium (he won)	Identical to the model's test-set prediction (assert)
Partially missing	Same, without qualifying gap, the 4 standings features, grid share and front row	P = 0.592 → podium	Grid share and front row derived exactly; gap filled with 0.93% (real 0.00%), standings with training medians (13th; real: leader)
Grid only	Grid slot 6, 20-car field	P = 0.114 → no podium	22 features imputed; pole gives only 0.33
Invalid input	`grid_rnk = 3`; `grid_rank = "P3"`	ValueError; TypeError	Typos are not silently treated as missing

With few known features the function is **conservative**: medians describe a midfield driver with almost no recent podiums, so front-runners are under-predicted (pole 0.33 vs about 63% observed). The `imputed` list tells the user how much was guessed.

12. Known limitations

- **Qualifying-coverage drift.** Qualifying data starts in 1994 and is complete only from 2003, so 63.7% of training rows have an imputed qualifying gap while validation and test are almost complete. Experiment 9.2 found no better alternative, but the training and deployment data differ.
- **Era sensitivity.** Field sizes, reliability and points systems changed across eras; the model fits pre-1994 seasons far worse (section 8.5). The test seasons are a new regulation era; performance held up, but a future rule change could behave differently.
- **Small evaluation sets.** Validation has 39 races and test 68, so PR-AUC differences of about 0.01–0.03 are within noise. All model comparisons are reported with race-level bootstrap intervals for this reason.
- **Race-day chaos is outside the cutoff.** Rain, safety cars and on-the-day failures cause most of the biggest errors (section 8.5). Wet qualifying also distorts the qualifying gap (capped at 7%).
- **Imputation at inference is conservative.** A record with few known features is predicted like an average midfield driver.

- **Sprint weekends.** Training contains no sprints, so the sprint result is unused even though it is pre-race and predictive.
- **Team identity resets.** Renamed teams get new ids, so team form and standings restart after a rename (bridged only by the driver-to-team fills).
- **Status taxonomy and home definition.** The mechanical vs non-mechanical grouping is hand-built (49 retirements without a cause), and nationality is a proxy for home.
- **Associations, not causes.** The DEQ answers and the model explanations describe patterns in the data; they do not establish why podiums happen.

13. Reproducibility and notebook map

The notebook runs top to bottom with a single Run All, locally or on Kaggle, and finds the data folder automatically. Random seeds are fixed. The XAI section installs `shap` and `lime` if missing (on Kaggle, turn Internet on). All figures in this report are produced by the notebook.

Notebook section	Content	Report section
1.1–1.14	Audit and cleaning of each table, with before/after EDA	2
2.1–2.5	Integrity audit: raw files unchanged, primary keys, foreign keys, cleaning impact, saved outputs	2
3.1–3.3	DEQ1–DEQ3 with sensitivity checks	3–5
4.0–4.10	Feature engineering by group, missing values, final model table and split	6
5.1–5.3	Preprocessing pipeline and preprocessing-order experiments	7
6.1–6.8	Models, comparison, bootstrap, error analysis	8
7.1–7.4	Experiments: ablation, qualifying handling, interaction feature, decision rule	9
8.1–8.3	Explainability: permutation importance, SHAP, local SHAP and LIME	10
9.1–9.4	Inference function and demos	11